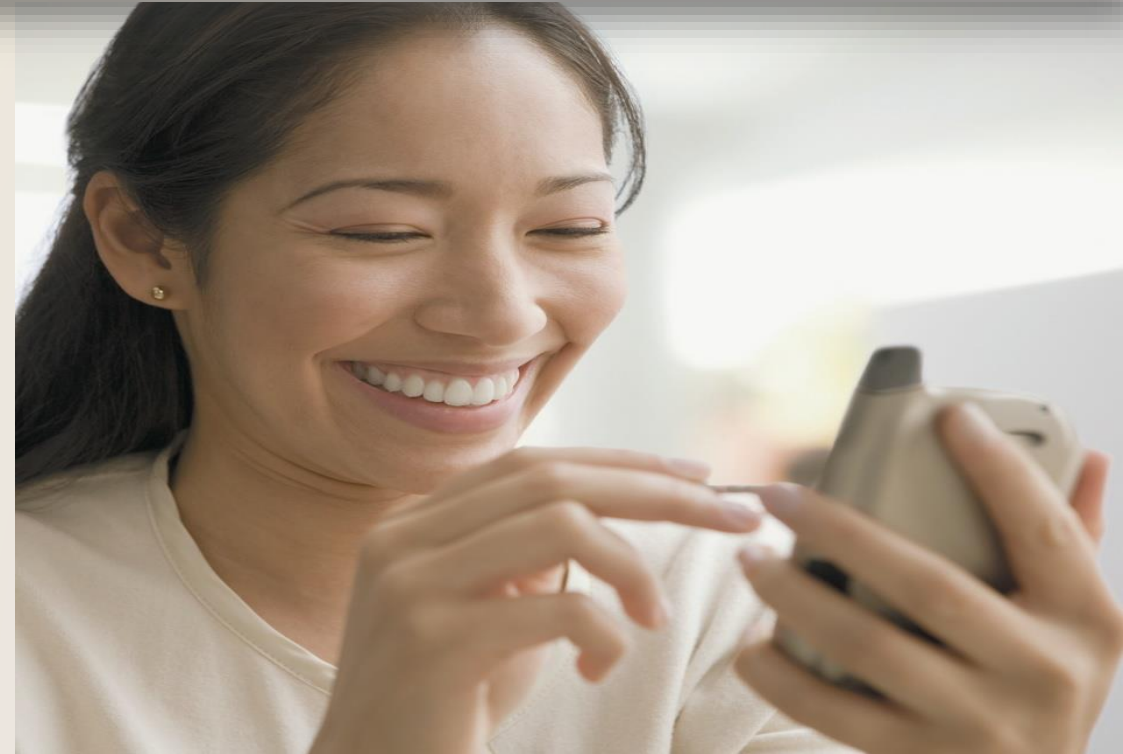


## Chapter Two

# The Internet and World Wide Web

## Discovering Computers 2012

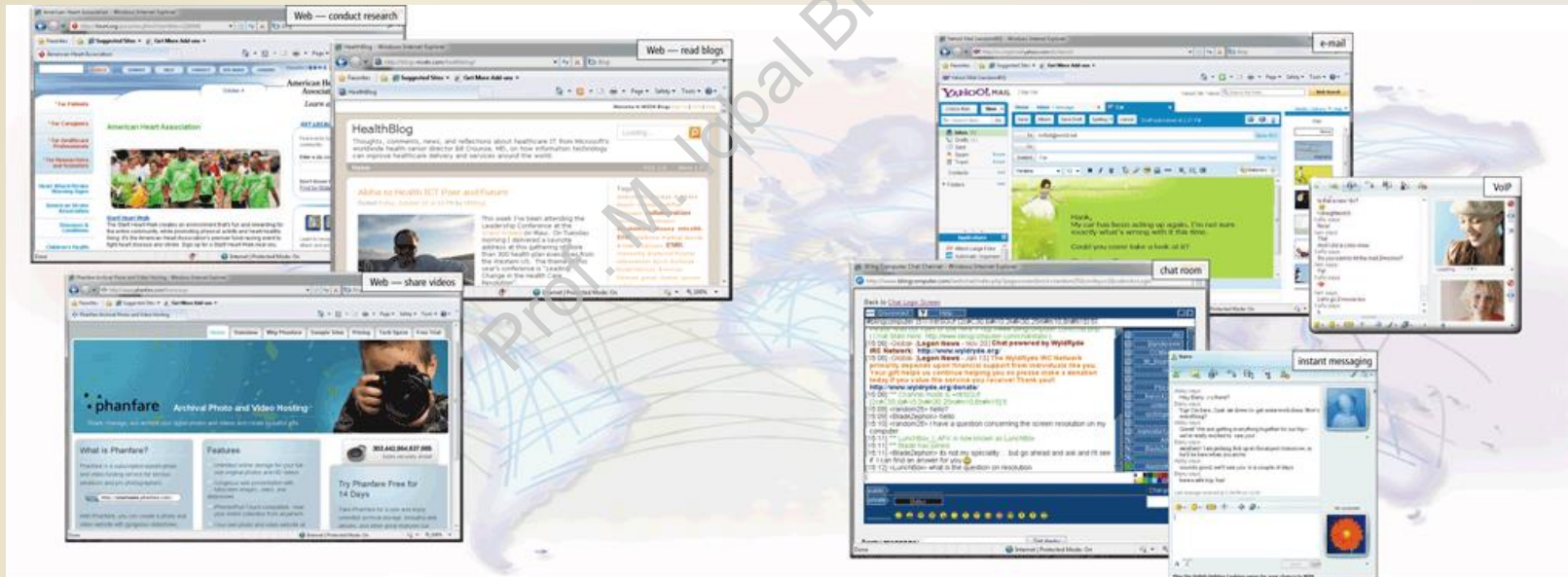
**Your Interactive Guide to the  
Digital World**



- The *Internet* has revolutionized many aspects of our daily lives. It has affected the way we do business as well as the way we spend our leisure time. The Internet is a communication system that has brought a wealth of information to our fingertips and organized it for our use.
- The Internet (or internet) is the global system of interconnected computer networks that uses the Internet protocol suite (TCP/IP) to communicate between networks and devices. It is a network of networks that consists of private, public, academic, business, and government networks of local to global scope, linked by a broad array of electronic, wireless, and optical networking technologies.

# The Internet

- The **Internet** is a worldwide collection of networks that links millions of businesses, government agencies, educational institutions, and individuals



# Evolution of the Internet

- The Internet originated as ARPANET in September 1969 and had two main goals:

Allow scientists at different physical locations to share information and work together

Function even if part of the network were disabled or destroyed by a disaster



## 27-1 ARCHITECTURE

*The idea of the Web was first proposed by Tim Berners-Lee in 1989 at CERN†, the European Organization for Nuclear Research, to allow several researchers at different locations throughout Europe to access each others' researches. The commercial Web started in the early 1990s.*

*The Web today is a repository of information in which the documents, called web pages, are distributed all over the world and related documents are linked together. The popularity and growth of the Web can be related to two terms in the above statement:*

- distributed and*
- linked.*

# The Early Days



## ARPANET

In the 1960s, the US government developed ARPANET, the first network that allowed computers to communicate with each other.



## Tim Berners-Lee

In 1991, British computer scientist Tim Berners-Lee created the World Wide Web, which transformed the internet into a global network.



## Dial-Up Modems

In the 1990s, dial-up modems became popular, allowing users to connect to the internet through a phone line.

# Invention of the Internet



In the 1960s, the United States government, businesses, and colleges collaborated to develop a system that would allow computers across the country to share information



In 1969, they developed ARPANET, an early form of the Internet



Electronic mail, or e-mail, was invented in 1971 as a method of sending a message from one computer to another

# Invention of the Internet



By the mid-1970s, many groups of computers had been linked together in networks



To connect the networks, machines known as routers were invented



This is how the first ARPANET evolved into the Internet

Prof. M. Iqbal Bhat (JKHED)



# Invention of the Internet



Hypertext was invented in 1989

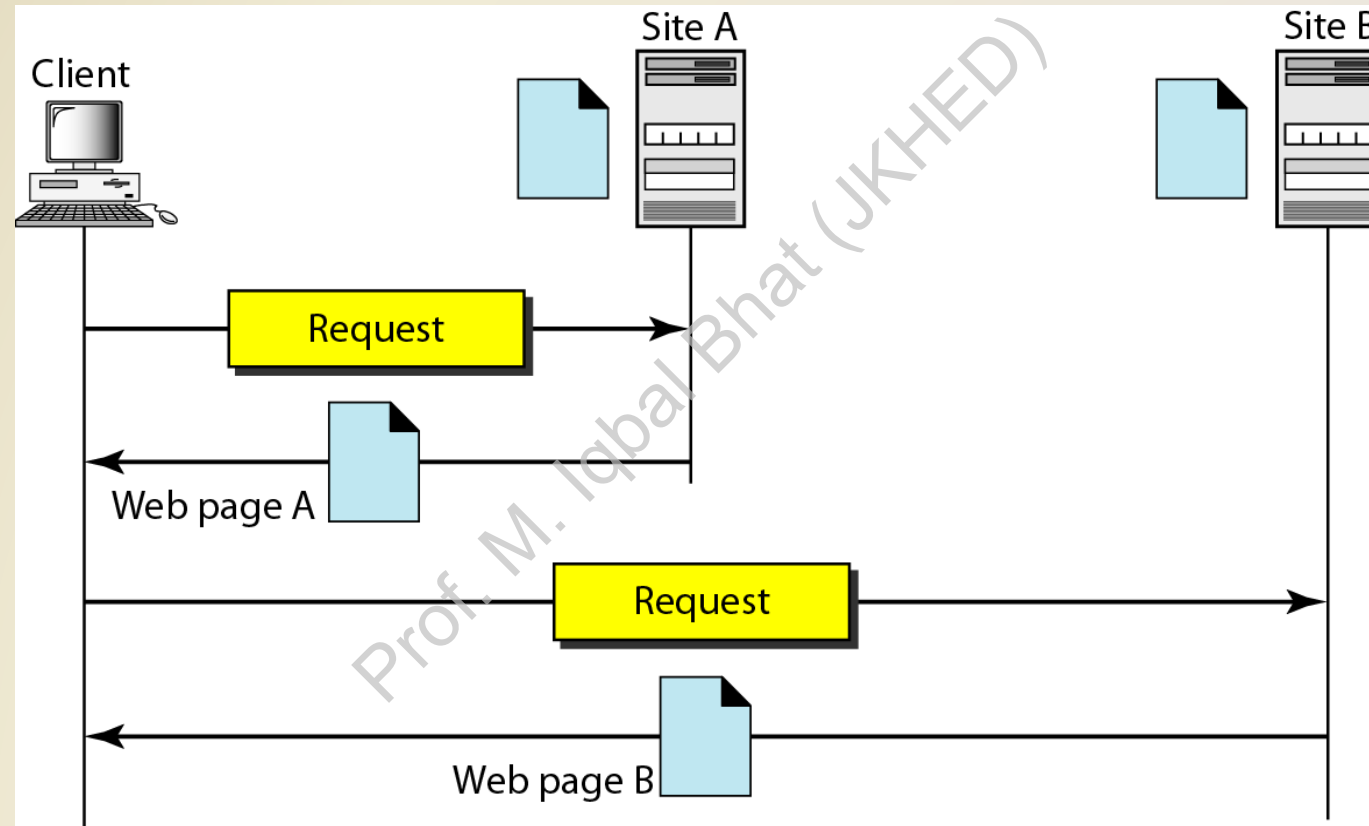


A hypertext link connects different parts of an electronic document or different documents



The World Wide Web, or "the Web," was created in the early 1990s on the basis of hypertext

## *Architecture of WWW: Client-Server Model*



# The Building Blocks

1

## **TCP/IP Protocol**

The TCP/IP protocol standardizes how data is transmitted over the internet.

2

## **DNS (Domain Name System)**

DNS translates domain names to IP addresses, enabling you to type in a website's name instead of its numerical address.

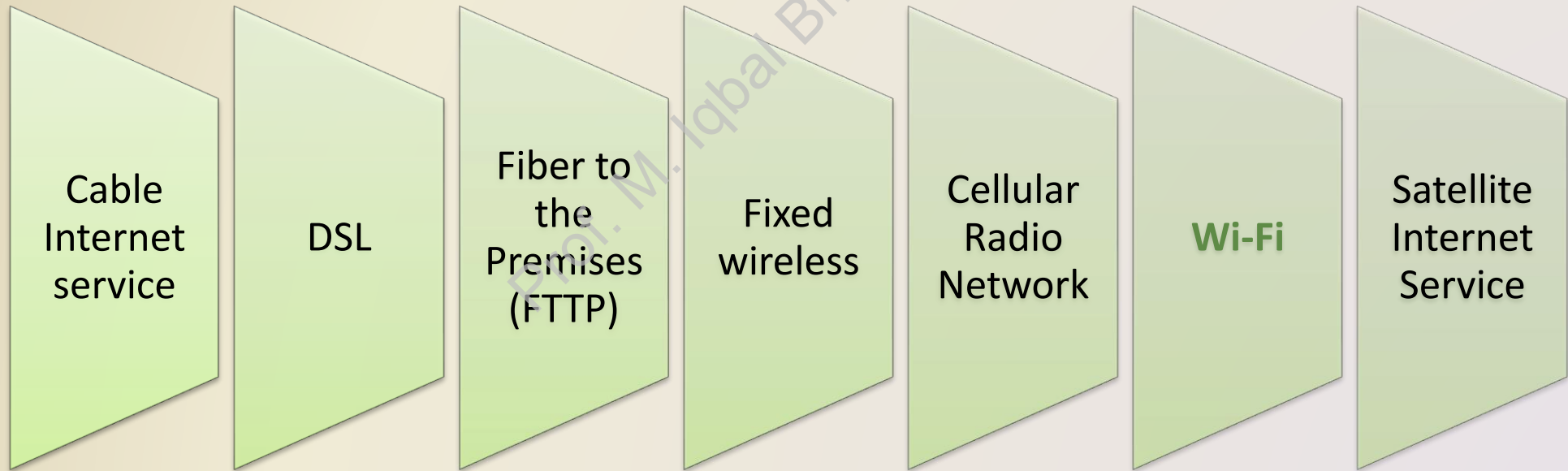
3

## **HTTP (Hypertext Transfer Protocol)**

HTTP governs how web pages are formatted and transmitted over the internet.

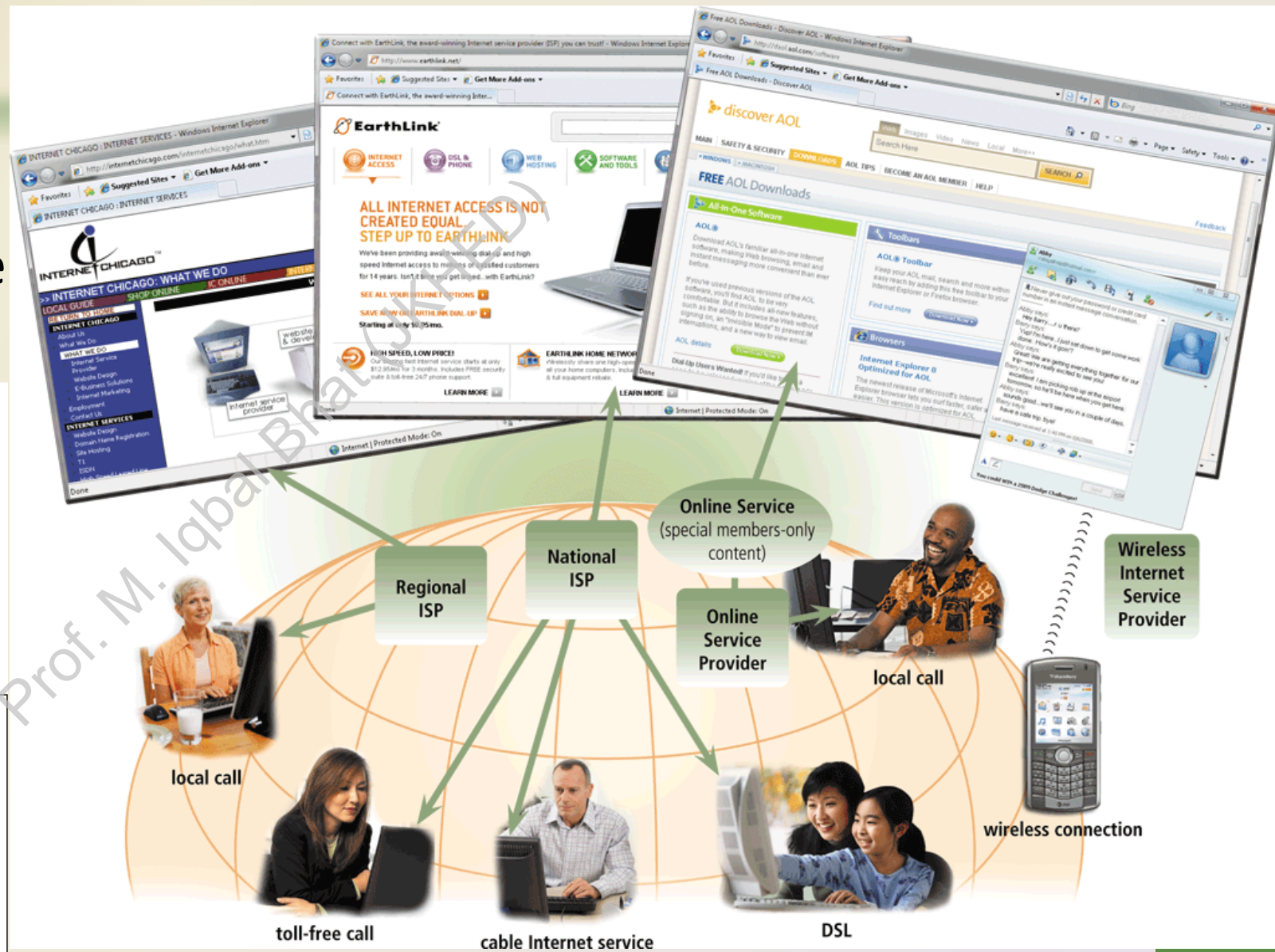
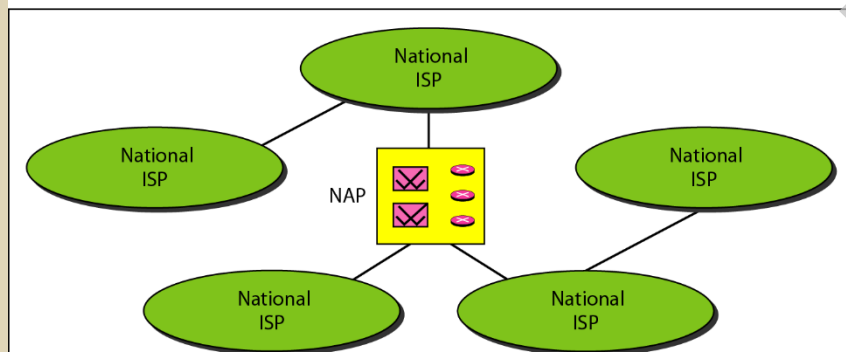
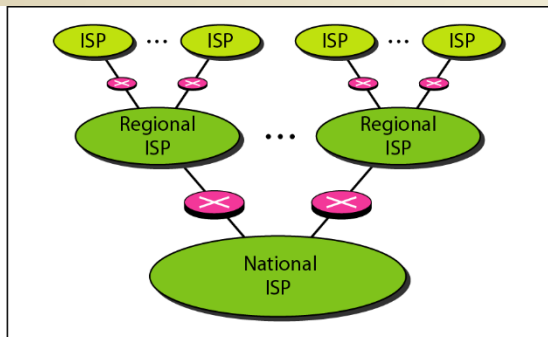
# Evolution of the Internet

- Many home and small business users connect to the Internet via high-speed broadband Internet service





- An **access provider** is a business that provides individuals and organizations access to the Internet free or for a fee



# Evolution of the Internet

## ISP (Internet service provider)

Regional ISPs provide Internet access to a specific geographical area

National ISPs provide Internet access in cities and towns nationwide

## Online service provider (OSP)

Has many members-only features

Popular OSPs include AOL (America Online) and MSN (Microsoft Network)

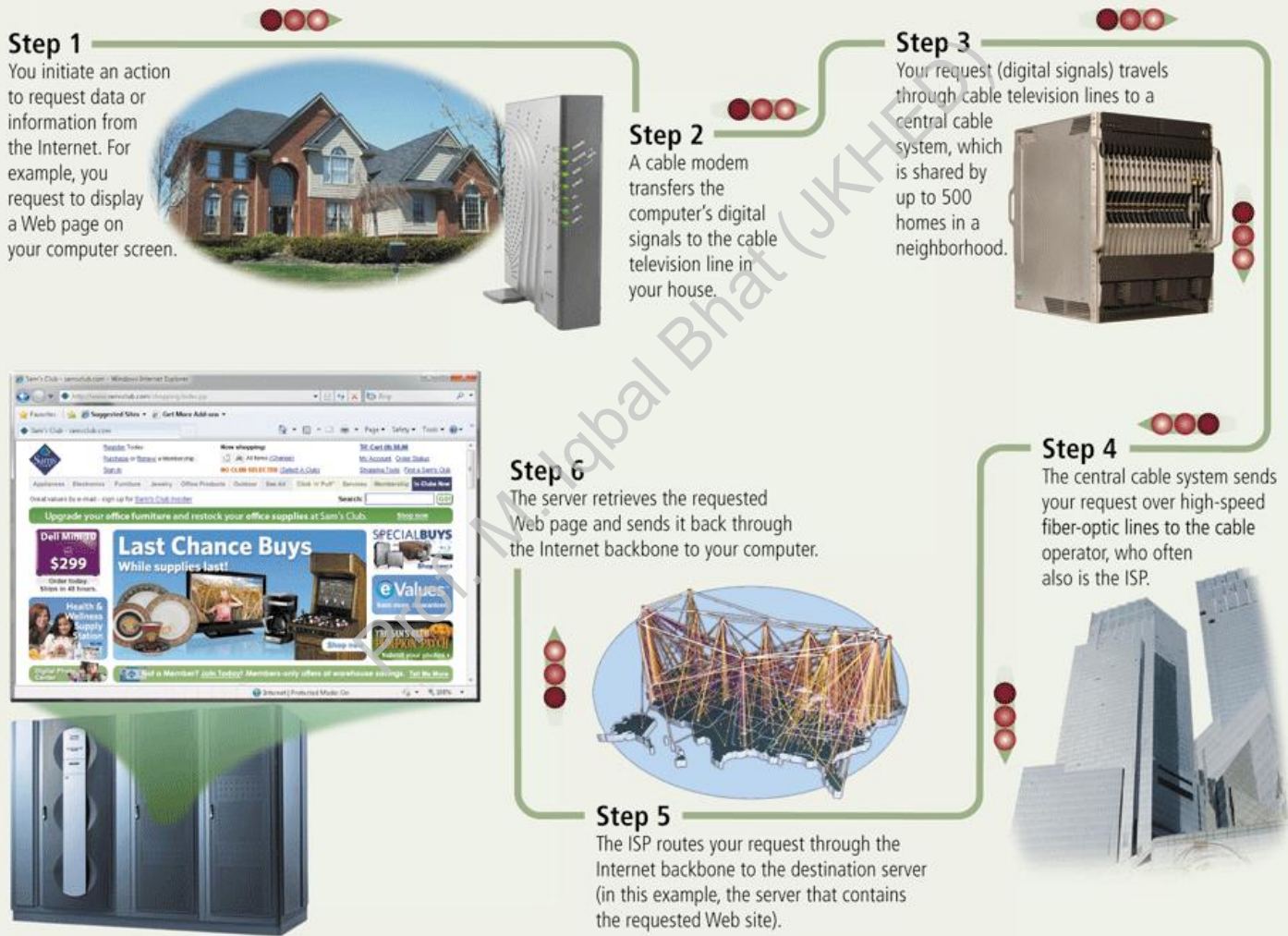
## Wireless Internet service provider

Provides wireless Internet access to computers and mobile devices

May require a wireless modem

# Evolution of the Internet

## How a Home User's Data and Information Might Travel the Internet Using a Cable Modem Connection





# Evolution of the Internet

- An **IP address** is a number that uniquely identifies each computer or device connected to the Internet
- A **domain name** is the text version of an IP address
  - Top-level domain (TLD)
- A **DNS server** translates the domain name into its associated IP address



# Evolution of the Internet

| Examples of Generic Top-Level Domains |                                                                       |
|---------------------------------------|-----------------------------------------------------------------------|
| Generic TLD                           | Intended Purpose                                                      |
| aero                                  | Aviation community members                                            |
| biz                                   | Businesses of all sizes                                               |
| cat                                   | Catalan cultural community                                            |
| com                                   | Commercial organizations, businesses, and companies                   |
| coop                                  | Business cooperatives such as credit unions and rural electric co-ops |
| edu                                   | Educational institutions                                              |
| gov                                   | Government agencies                                                   |
| info                                  | Business organizations or individuals providing general information   |
| jobs                                  | Employment or human resource businesses                               |
| mil                                   | Military organizations                                                |
| mobi                                  | Delivery and management of mobile Internet services                   |
| museum                                | Accredited museums                                                    |
| name                                  | Individuals or families                                               |
| net                                   | Network providers or commercial companies                             |
| org                                   | Nonprofit organizations                                               |
| pro                                   | Certified professionals such as doctors, lawyers, and accountants     |
| tel                                   | Internet communications                                               |
| travel                                | Travel industry                                                       |



# The World Wide Web

- The **World Wide Web**, or **Web**, consists of a worldwide collection of electronic documents (**Web pages**)
- A **Web site** is a collection of related Web pages and associated items
- A **Web server** is a computer that delivers requested Web pages to your computer

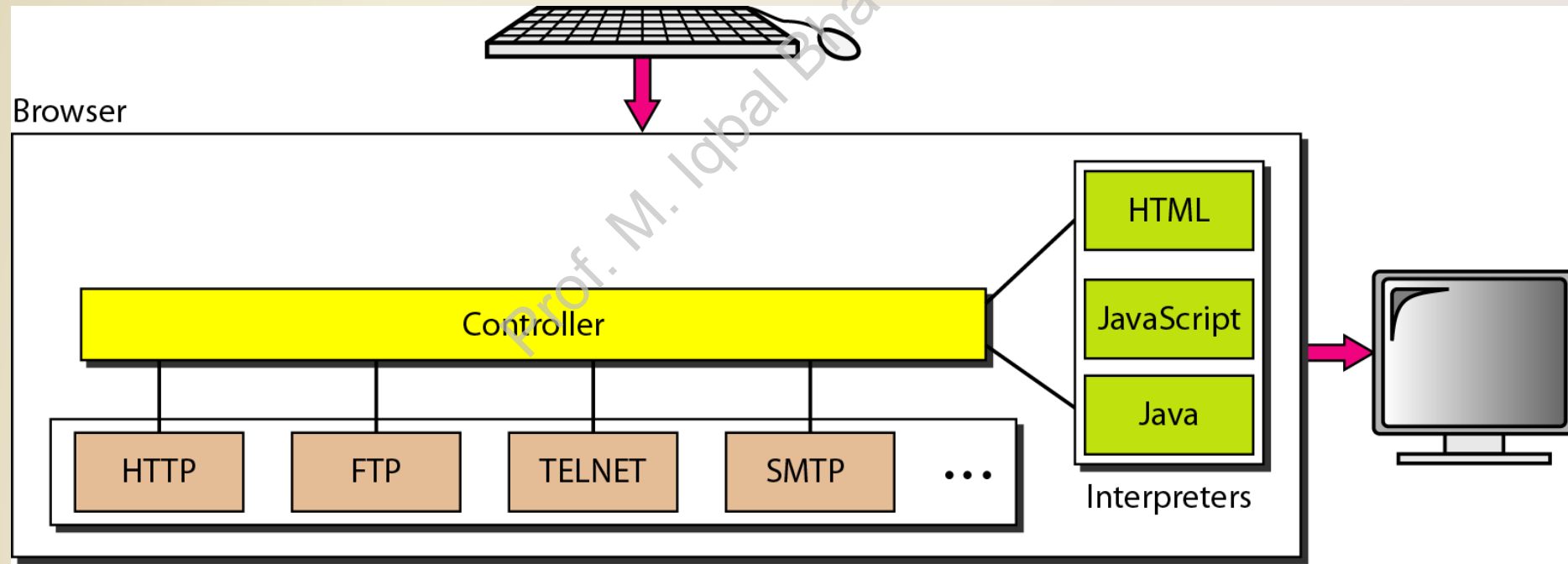
# Web Browser:

- A web browser is a software application that enables users to access, retrieve, and view information on the World Wide Web. It acts as a gateway to the internet, allowing users to interact with various online resources such as websites, images, videos, and other content.



# Browsers:

Each browser usually consists of three parts: a controller, client protocols, and interpreters.



# The World Wide Web

- A Web page has a unique address called a **URL** or **Web address**





---

# URL

---

*A URL, or Uniform Resource Locator, is a web address that is used to identify and locate resources on the internet. It serves as a reference or an address for resources such as web pages, images, files, and other types of content.*

*A web page, as a file, needs to have a unique identifier to distinguish it from other web pages. To define a web page, we need three identifiers: host, port, and path.*



**protocol://host/path**

**Used most of the time**

**protocol://host:port/path**

**Used when port number is needed**

# The World Wide Web

- A **home page** is the first page that a Web site displays
- Web pages provide **links** to other related Web pages
  - **Surfing the Web**
- **Downloading** is the process of receiving information
- Some Web pages are designed specifically for microbrowsers



# The World Wide Web

- **Tabbed browsing** allows you to open and view multiple Web pages in a single Web browser window



# Search Engine:

- Two types of search tools are search engines and subject directories

## Search engine

A search engine is a software program or an online service that enables users to search for information on the internet. The primary function of a search engine is to help users discover relevant content based on their queries or keywords. Search engines use complex algorithms to index and rank vast amounts of web pages, making it easier for users to find the information they are looking for.

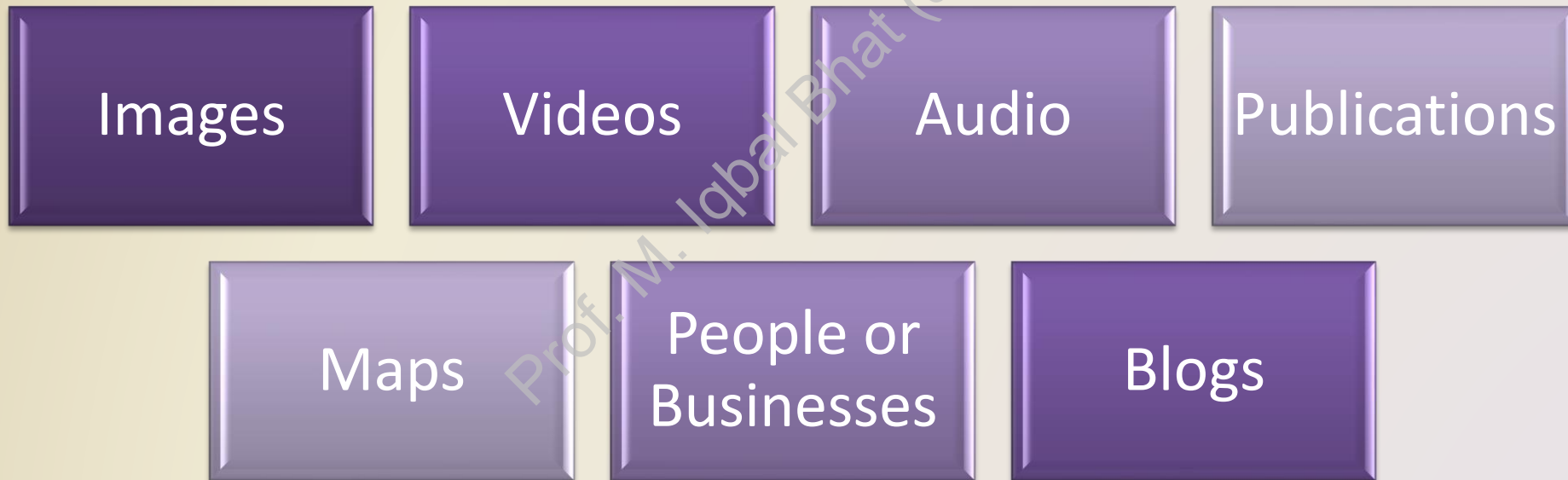


## Subject directory

Classifies Web pages in an organized set of categories

# The World Wide Web

- A search engine is helpful in locating items such as:





# Key components and functions of a search engine include:

- **Crawling**: Search engines employ automated bots, often called spiders or crawlers, to systematically browse and index web pages across the internet. During this process, the crawler follows links from one page to another, collecting information about the content on each page.
- **Indexing**: The collected information is then organized and stored in a searchable index. This index contains data about the keywords, content, and structure of web pages. Indexing allows search engines to quickly retrieve relevant information when responding to user queries.
- **Ranking**: When a user enters a search query, the search engine uses its algorithm to evaluate the relevance of indexed pages and ranks them based on various factors. These factors may include keyword relevance, page quality, user experience, and the number of links pointing to a page.
- **Search Query Processing**: Users input their queries into the search engine's search bar, and the search engine processes these queries to identify relevant results from its index.
- **Results Display**: Search engines present the results in the form of a Search Engine Results Page (SERP). The SERP typically includes a list of links to web pages, along with brief descriptions (snippets) and other features like images, videos, or featured snippets.

# Websites:

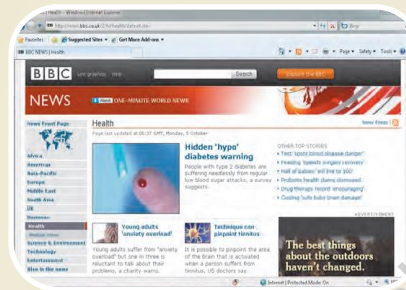
- A website is a collection of web pages and related content that is identified by a common domain name and published on at least one web server. It is accessible through the internet or a private local area network (LAN) using a web browser. Websites are designed and created for various purposes, including information dissemination, communication, e-commerce, entertainment, and more.

# Websites

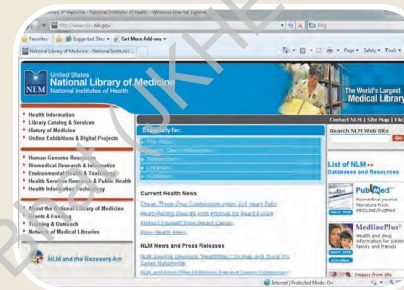
- There are thirteen types of Web sites



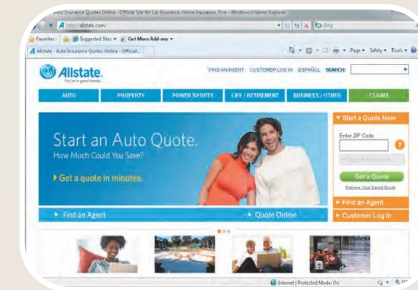
Portal



News



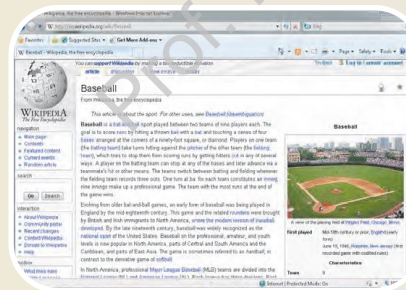
Informational



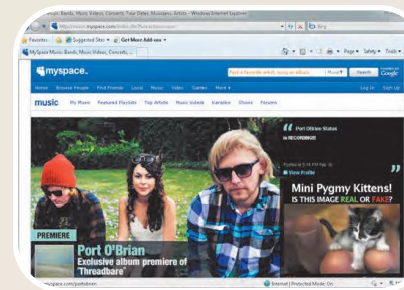
Business/Marketing



Blog



Wiki



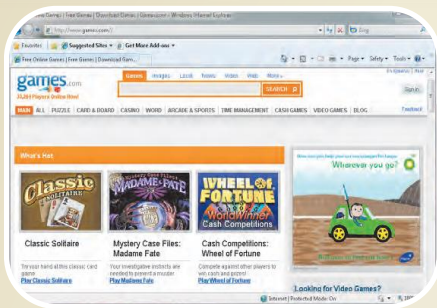
Online Social Network



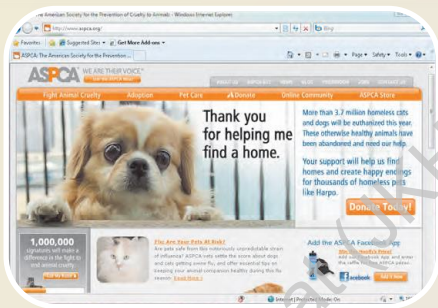
Educational



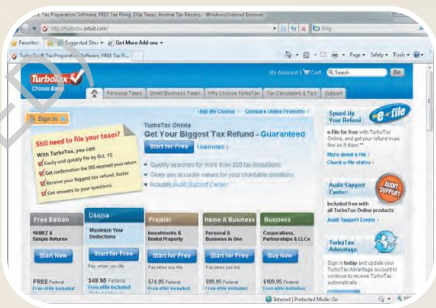
# The World Wide Web



Entertainment



Advocacy



Web Application



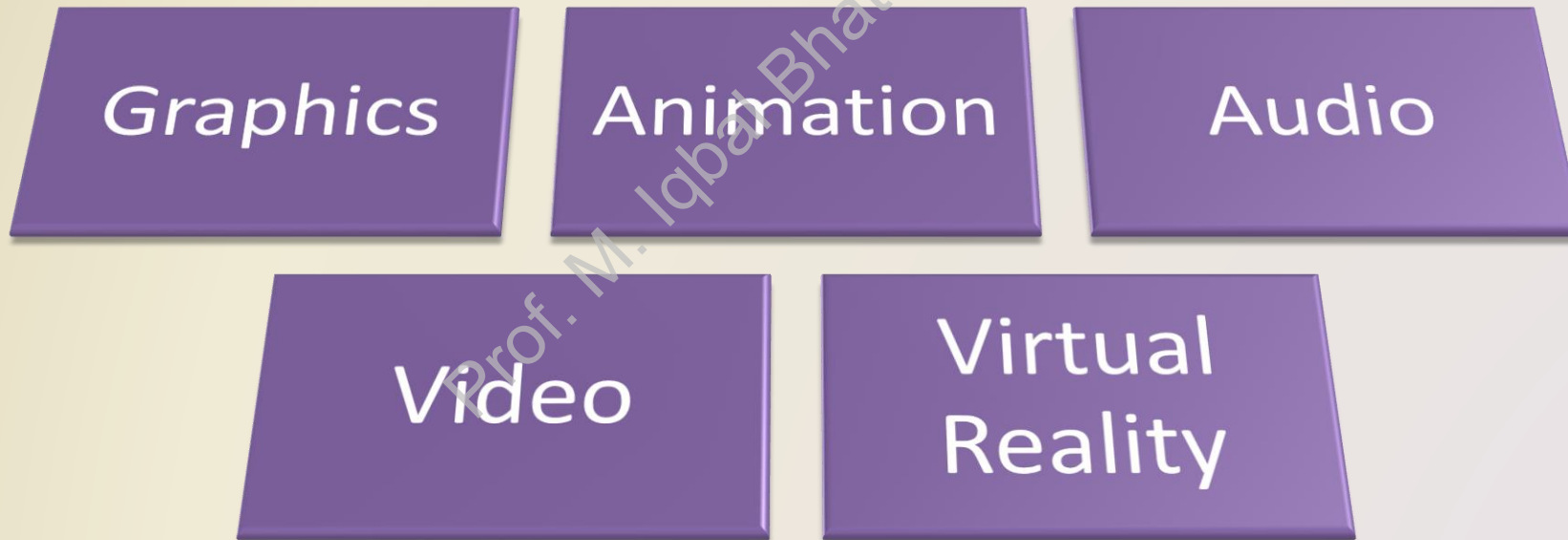
Content Aggregator



Personal

# What is Multimedia:

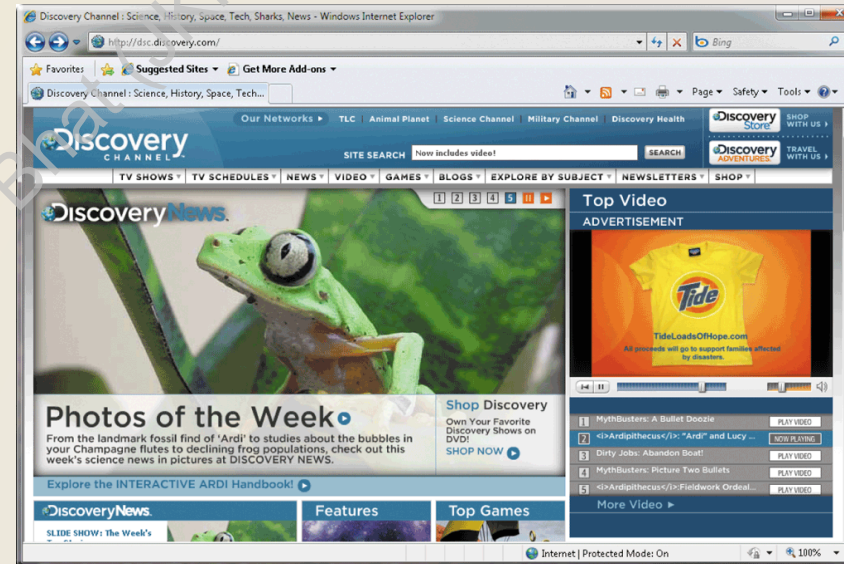
- **Multimedia** refers to any application that combines text with:





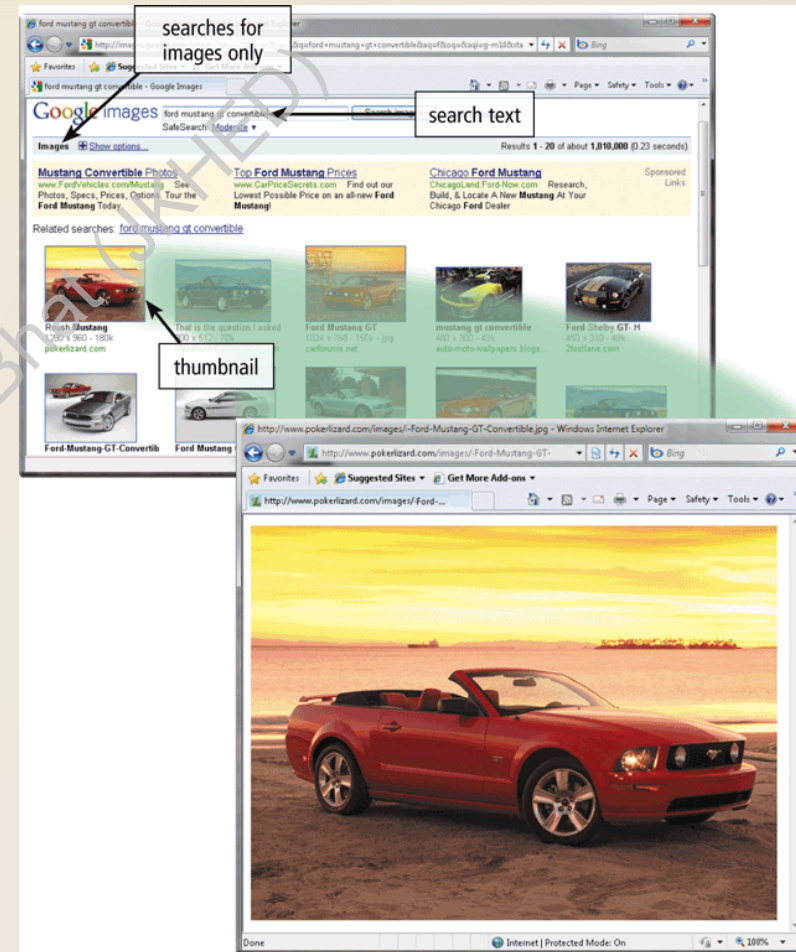
# The World Wide Web

- A **graphic** is a digital representation of non text information
- Graphic formats include BMP, GIF, JPEG, PNG, and TIFF



# The World Wide Web

- A thumbnail is a small version of a larger graphic



# The World Wide Web

- **Animation** is the appearance of motion created by displaying a series of still images in sequence



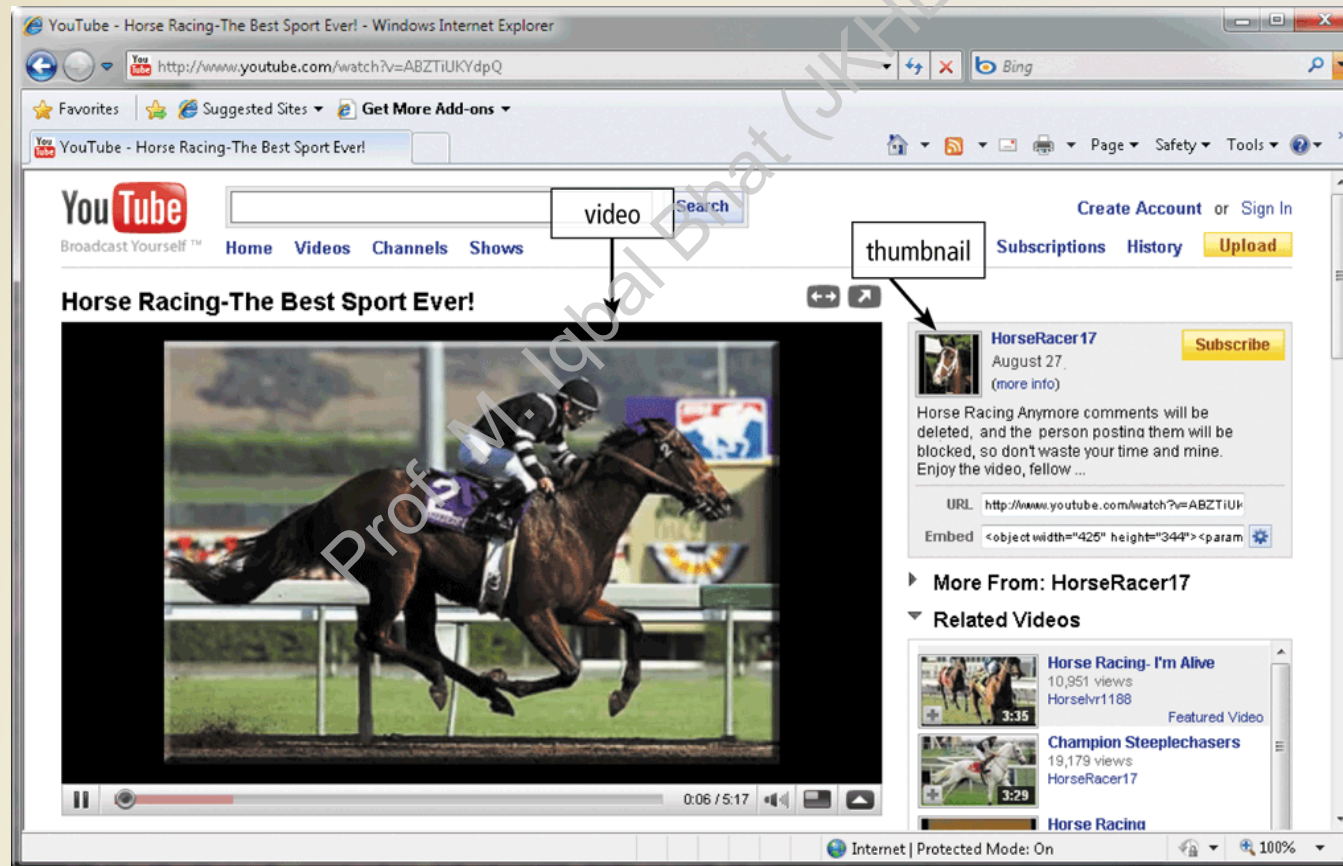
# The World Wide Web

- **Audio** includes music, speech, or any other sound
  - Compressed to reduce file size
- You listen to audio on your computer using a **player**
- **Streaming** is the process of transferring data in a continuous and even flow

| Audio Web File Formats |                                                                             |        |                                                          |
|------------------------|-----------------------------------------------------------------------------|--------|----------------------------------------------------------|
| Format                 | Description                                                                 | Format | Description                                              |
| AAC                    | Advanced Audio Coding                                                       | WAV    | Windows waveform                                         |
| AIFF                   | Audio Interchange File Format                                               | WMA    | Windows Media Audio<br>(part of Windows Media framework) |
| ASF                    | Advanced Streaming (or Systems) Format<br>(part of Windows Media framework) | RA     | RealAudio sound file<br>(supported by RealPlayer)        |
| MP3                    | Moving Pictures Experts Group Audio Layer 3<br>(MPEG-3)                     | QT     | QuickTime audio, video, or 3-D animation                 |
| Ogg                    | Free, unpatented audio and video format                                     |        |                                                          |

# The World Wide Web

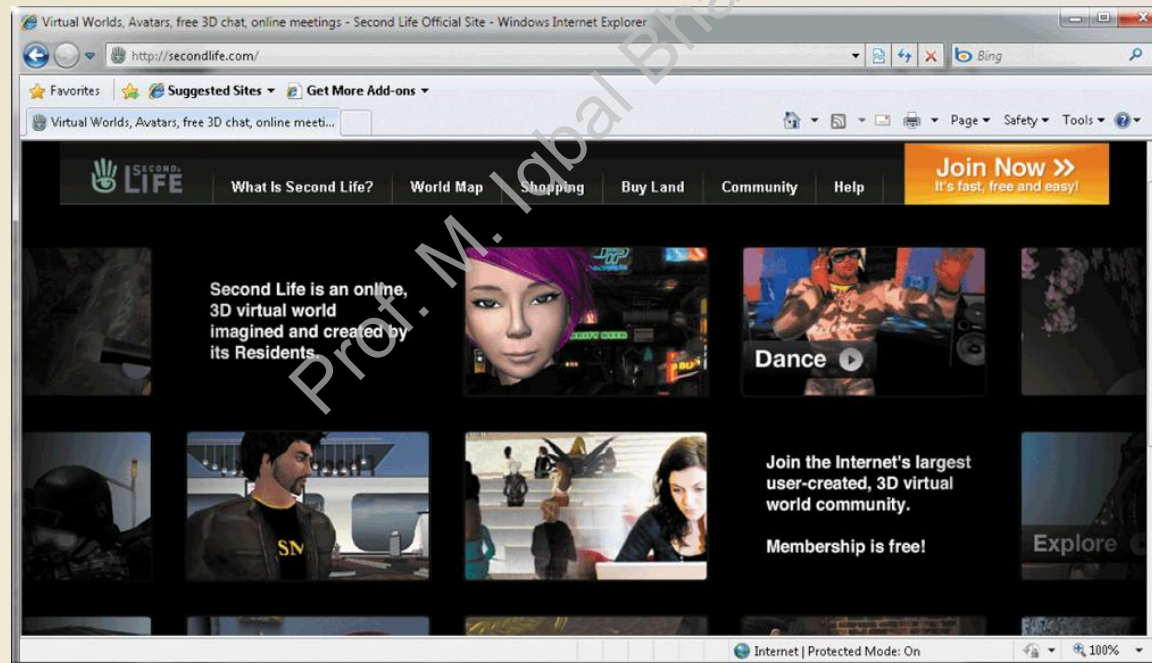
- **Video** consists of images displayed in motion





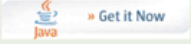
# The World Wide Web

- **Virtual reality (VR)** is the use of computers to simulate a real or imagined environment that appears as a three-dimensional space



# The World Wide Web

- A **plug-in** is a program that extends the capability of a Web browser

| Popular Plug-Ins     |                                                                                     |                                                                                                                                                                    |               |
|----------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Plug-In Application  |                                                                                     | Description                                                                                                                                                        | Web Address   |
| Acrobat Reader       |    | View, navigate, and print Portable Document Format (PDF) files — documents formatted to look just as they look in print                                            | adobe.com     |
| Flash Player         |    | View dazzling graphics and animation, hear outstanding sound and music, display Web pages across an entire screen                                                  | adobe.com     |
| Java                 |    | Enable Web browser to run programs written in Java, which add interactivity to Web pages                                                                           | java.com      |
| QuickTime            |    | View animation, music, audio, video, and VR panoramas and objects directly on a Web page                                                                           | apple.com     |
| RealPlayer           |   | Listen to live and on-demand near-CD-quality audio and newscast-quality video, stream audio and video content for faster viewing, play MP3 files, create music CDs | real.com      |
| Shockwave Player     |  | Experience dynamic interactive multimedia, 3-D graphics, and streaming audio                                                                                       | adobe.com     |
| Silverlight          |  | Experience high-definition video, high-resolution interactive multimedia, and streaming audio and video                                                            | microsoft.com |
| Windows Media Player |  | Listen to live and on-demand audio, play or edit WMA and MP3 files, burn CDs, and watch DVD movies                                                                 | microsoft.com |

# The World Wide Web

- **Web publishing** is the development and maintenance of Web pages



# E-Commerce

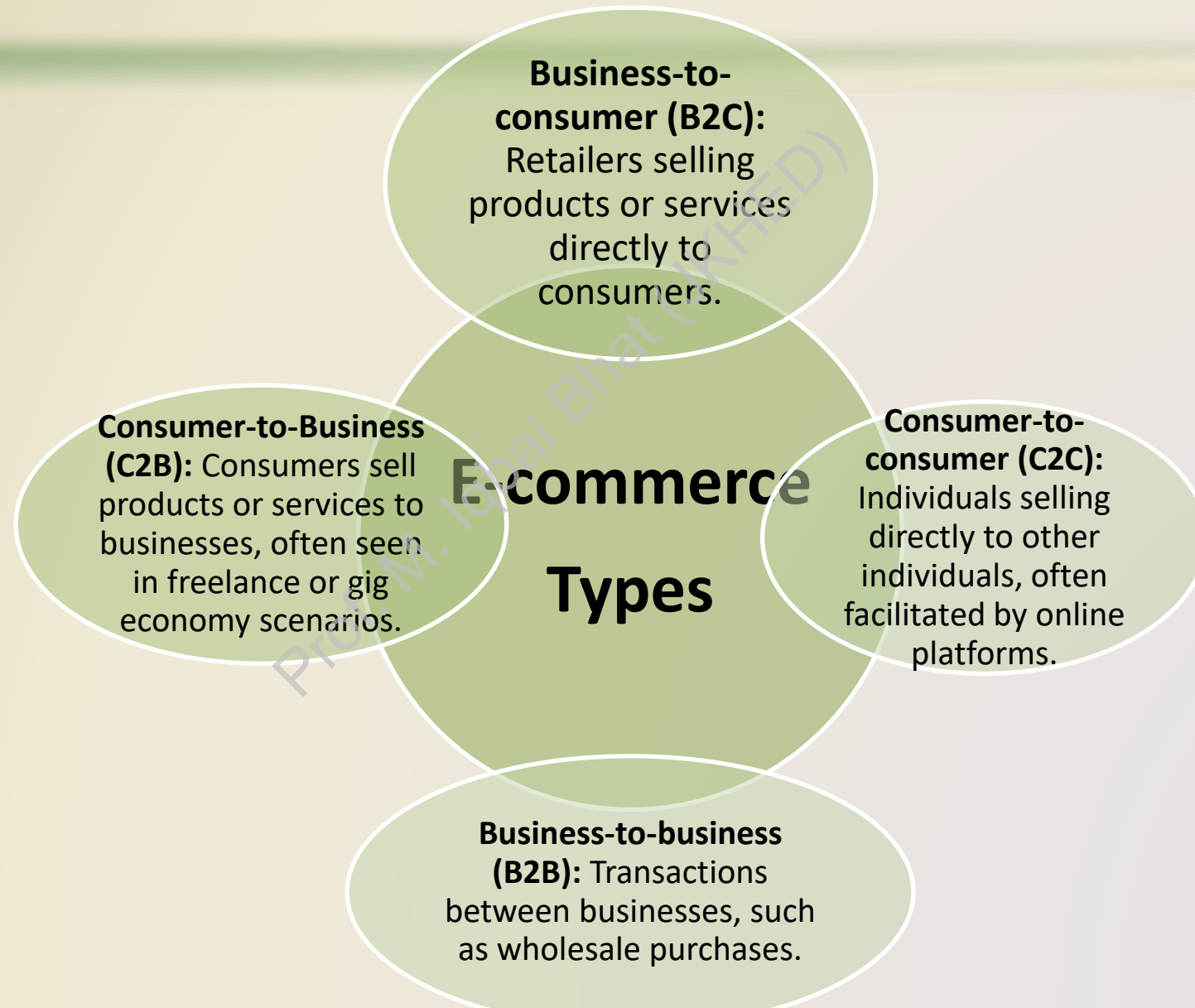
- **E-commerce** short for electronic commerce, refers to the buying and selling of goods and services over the internet. It involves online transactions between businesses and consumers, businesses and businesses, and consumers and consumers. E-commerce has become a significant aspect of the global economy, transforming the way people shop, conduct business, and exchange value.

# E-Commerce Components:

- **Online Transactions**: E-commerce enables the exchange of goods, services, and money through online transactions. Customers can browse products or services on websites, place orders, and make payments electronically.
- **Digital Storefronts**: E-commerce businesses typically have digital storefronts or online platforms where they showcase their products or services. These storefronts can range from simple websites to sophisticated e-commerce platforms.
- **Electronic Payments**: E-commerce transactions involve electronic payment methods. Customers can make payments using credit cards, digital wallets, online banking, and other electronic payment systems.
- **Online Shopping**: Consumers can browse product catalogs, view detailed product information, compare prices, and make purchase decisions online. The entire shopping experience, from product discovery to payment, occurs through digital channels.
- **Global Reach**: E-commerce has a global reach, allowing businesses to connect with customers beyond geographical boundaries. This global accessibility expands market opportunities for businesses and provides consumers with a wide array of choices.
- **Security Measures**: Due to the sensitive nature of online transactions, e-commerce platforms implement robust security measures to protect customer information and ensure secure financial transactions. This often involves encryption and secure payment gateways.
- **Logistics and Fulfillment**: E-commerce businesses need efficient logistics and fulfillment processes to handle order processing, packaging, and delivery. This includes coordinating with shipping carriers for timely and reliable delivery of products.



# E-Commerce Types:



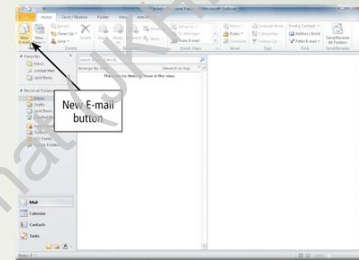
# Other Internet Services

- **E-mail** is the transmission of messages and files via a computer network
- An **e-mail program** allows you to create, send, receive, forward, store, print, and delete e-mail messages

## How to Send an E-Mail Message Using Outlook

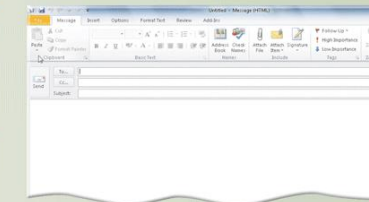
### Step 1

Start an e-mail program and point to the New Mail Message button.



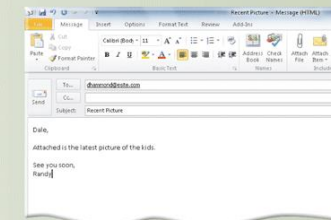
### Step 2

Click the New Mail Message button to display the Message window.



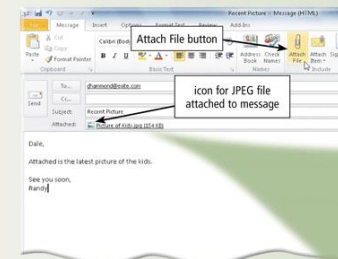
### Step 3

Enter the recipient's e-mail address, the subject, and the message in the Message window.



### Step 4

Click the Attach File button on the Message tab to attach a JPEG file containing a photo to the message. Click the Send button to send the message.



### Step 5

When Dale receives the e-mail message, she opens the JPEG file to view the photo.

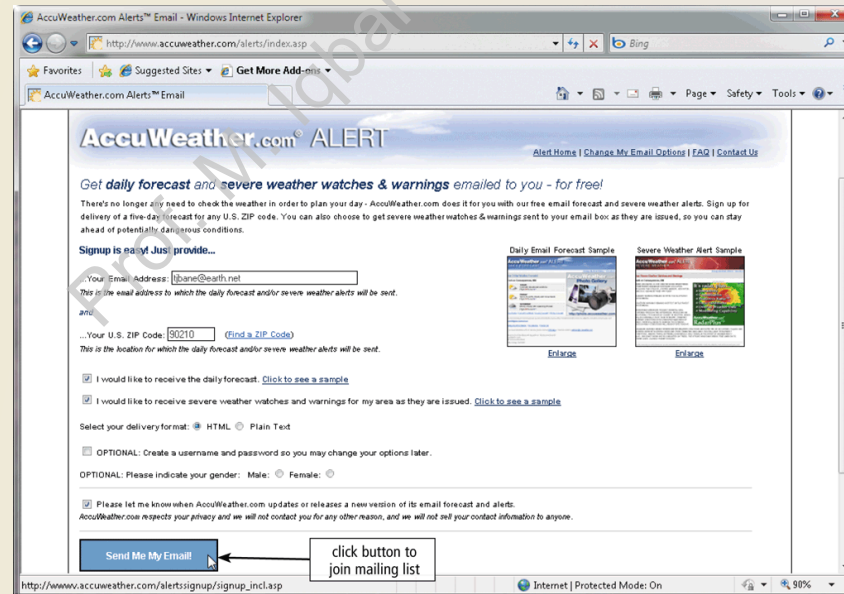


# Other Internet Services



# Other Internet Services

- A **mailing list** is a group of e-mail names and addresses given a single name
  - **Subscribing** adds your e-mail name and address
  - **Unsubscribing** removes your name



# Other Internet Services

- **Instant messaging (IM)** is a **real-time** Internet communications service





# Other Internet Services

- A **chat** is a real-time typed conversation that takes place on a computer
- A **chat room** is a location on an Internet server that permits users to chat with each other



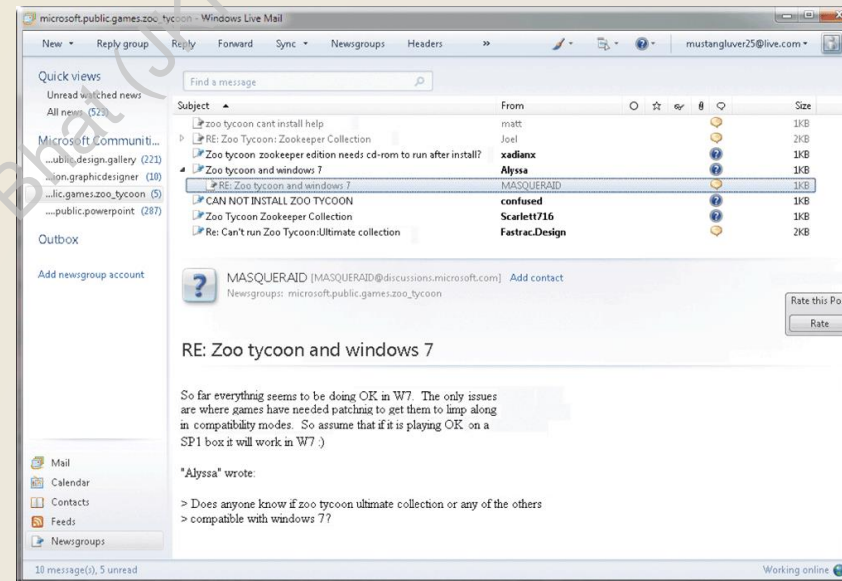
# Other Internet Services

- **VoIP** (Voice over IP) enables users to speak to other users over the Internet
  - Also called Internet telephony



# Other Internet Services

- A **newsgroup** is an online area in which users have written discussions about a particular subject
  - Typically requires a newsreader
- A **message board** is a Web-based type of discussion group



# Netiquette

- **Netiquette** refers to the set of rules and guidelines governing respectful and courteous behavior on the internet. It encompasses the manners and norms that individuals should follow when communicating and interacting online. Practicing netiquette contributes to positive and constructive online environments, whether in emails, social media, discussion forums, or other online platforms.

NETIQUETTE — Golden Rule: Treat others as you would like them to treat you

1. In e-mail, chat rooms, and newsgroups:

- Keep messages brief. Use proper grammar, spelling, and punctuation.
- Be careful when using sarcasm and humor, as it might be misinterpreted.
- Be polite. Avoid offensive language.
- Read the message before you send it.
- Use meaningful subject lines.
- Avoid sending or posting *flames*, which are abusive or insulting messages. Do not participate in *flame wars*, which are exchanges of flames.
- Avoid sending spam, which is the Internet's version of junk mail. *Spam* is an unsolicited e-mail message or newsgroup posting sent to many recipients or newsgroups at once.
- Do not use all capital letters, which is the equivalent of SHOUTING!
- Use **emoticons** to express emotion. Popular emoticons include
  - :) Smile     :| Indifference     :o Surprised
  - :( Frown     :\ Undecided

- Use abbreviations and acronyms for phrases:
  - btw     by the way
  - imho     in my humble opinion
  - fyi     for your information
  - ttfn     ta ta for now
  - fwiw     for what it's worth
  - tyvm     thank you very much
- Clearly identify a *spoiler*, which is a message that reveals a solution to a game or ending to a movie or program.

2. Read the *FAQ* (frequently asked questions), if one exists. Many newsgroups and Web pages have an *FAQ*.

3. Do not assume material is accurate or up-to-date. Be forgiving of other's mistakes.

4. Never read someone's private e-mail.

Page 108  
Figure 2-33

50

# Netiquette

- **Be Respectful::**
  - Treat others with the same respect and courtesy you would in face-to-face communication.
  - Avoid offensive language, insults, or personal attacks.
- **Use Proper Language:**
  - Avoid using all capital letters (which can be perceived as shouting).
  - Be mindful of grammar and spelling to enhance clarity and professionalism.
- **Mindful Posting:**
  - Think before you post, comment, or share content.
  - Consider the potential impact of your words on others.
- **Stay on Topic:**
  - Keep discussions relevant to the topic at hand.
  - Avoid derailing conversations with unrelated or off-topic comments.
- **Be Concise:**
  - Keep your messages concise and to the point.
  - Long, rambling messages may be less likely to be read thoroughly.



# Netiquette

- **Respect Privacy:**
  - Respect the privacy of others by not sharing personal information without consent.
  - Be cautious about sharing your own personal information.
- **Use Emoticons Wisely:**
  - Emoticons and emojis can help convey tone in text-based communication, but use them judiciously.
  - Avoid excessive use that may be distracting or unprofessional.
- **Understand the Platform's Culture:**
  - Different online platforms may have distinct cultures and norms.
  - Familiarize yourself with the specific rules and expectations of each platform.
- **Respond Thoughtfully:**
  - When responding to messages or comments, take the time to read and understand before replying.
  - Avoid knee-jerk reactions that can lead to misunderstandings.
- **Be Tolerant and Inclusive:**
  - Recognize and respect diverse perspectives and backgrounds.
  - Avoid discriminatory language or behavior.
- **Acknowledge Mistakes:**
  - If you make a mistake, acknowledge it and, if necessary, apologize.
  - Learn from any missteps to improve your online interactions.



# Thank You

Prof. M. Iqbal Phat (JKHED)